

Different types of Network measurements & $\Rightarrow$

The characterization & classification of natural and human-made structures using complex networks investigate the same important question of how to choose the most important appropriate measurements.

There are four main types of complex networks: —

- ① Weighted digraphs (directed graphs)
- ② Unweighted digraphs

Types of N/W measurements: $\Rightarrow$

- ① Active measurements
- ② Flow measurements
- ③ Packet measurements
- ④ Traffic measurements

end to end performance

Average download time of a web page

TCP bulk throughput

end-to-end delay & loss

state

Active routes

Active topology

Link bit error rate

Link utilization

traffic matrix

July															
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa		
13	14	15	16	17	18	19	20	21	22	23	24	25	26		
27	28	29	30	31											

mostly In Data Networks, the active, passive and hybrid measurements are introduced.

(a) Active measurements: →

This measurements generate special probe packets that are sent over the NW to, for eg., measure the time it takes for the packet to reach the other end of the NW (one way delay) the available capacity of a NW path or response time of an application.

Unlike passive measurements, active measurements generate additional NW traffic so they may possibly disturb the normal traffic flow.

This is the case in most SLA-measurements where the measurement is done constantly meaning the test traffic & customer traffic shares the same bandwidth.

(b) Methods for network measurements:-

- (a) Layer 2 measurement
- (b) Layer 2 + to Layer 4 measurements

These methods are used for measuring the available bandwidth on a path

August 2025													
Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	
11	12	13		1	2	3	4	5	6	7	8	9	
14	15	16	17	18	19	20	21	22	23				
24	25	26	27	28	29	30	31						

→ Traffic

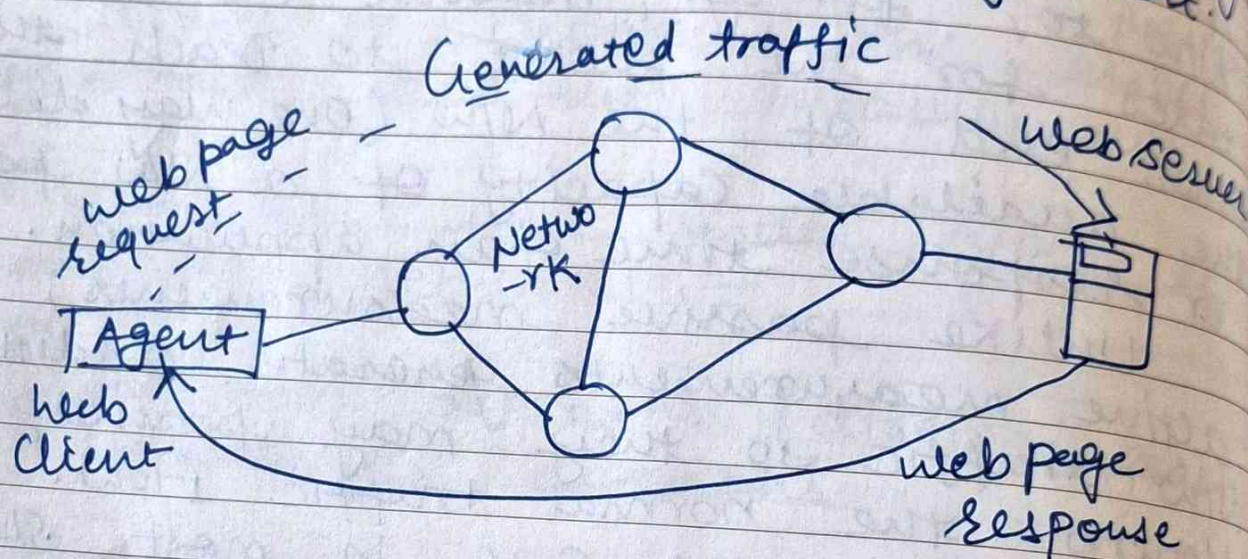
(2)

25

Friday • WK 30 (206-159)

Consist in sending probe packets at an increasing rate and recording the rate at which the probe's delays start to rise.

These methods will cause perturbations in the normal traffic flow. Although the perturbations are usually short.



An example of active NIW measurements

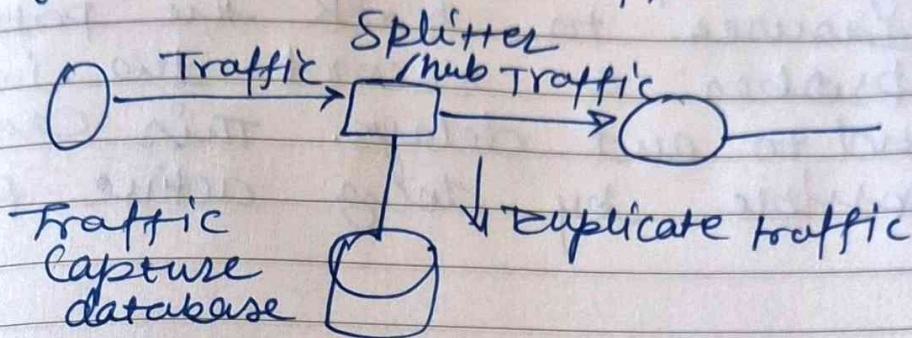
This measurements do not require huge amounts of storage space and they can be used to measure things that are not possible by using passive measurements.

July

Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su
			1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
27	28	29	30	31										

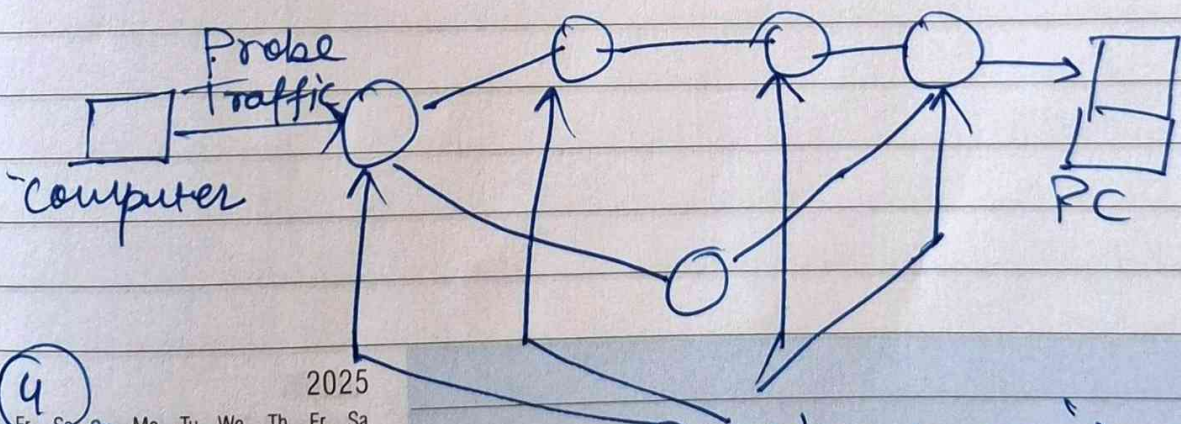
B. Passive N/W measurements ! → In this

N/W meas., the data is gathered passively listening to N/W traffic for example by using link splitters or hubs to duplicate a link's traffic or by monitoring buffers in routers.



An example of N/W measurement

① Hybrid measurement ! → Combining both active and passive measurements are called hybrid



an example of Hybrid N/W meas.

2025

Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27
28	29	30	31					

27

Sunday • WK 30 (208-157)

the above example is an example of
 09 hybrid measurements that could be a
 scenario where active probes are sent
 10 over a new and their progress
 is monitored by passive means
 11 during the measurement.
 This type of arrangement allows
 12 the measurees to track the path of
 the probes & record the intermediate
 01 and end to end delays. This can not
 be possible by doing active probing.

July

Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
		1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24	25	26
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